

Virtual Day: a VR Game for Evaluation of Prospective Memory in Older Adults

Maryam Alimardani¹, Guido Morera², Alexandra Hering²

¹ Vrije Universiteit Amsterdam, ² Tilburg University

ABSTRACT

Prospective memory (PM), the ability to remember to perform tasks in the future, is vital for maintaining functional independence in older adults. This paper introduces “Virtual Day”, a novel Virtual Reality (VR) game designed to assess PM in an immersive and realistic environment. We report a feasibility study where Virtual Day was compared to its clinical counterpart, a digital board game. Results indicate a preference for Virtual Day, noting it as more engaging and enjoyable, despite being more challenging. Follow-up development will build on these findings to use VR as a potential tool for cognitive interventions to promote healthy aging.

Keywords: Healthy aging, prospective memory, virtual reality

1 INTRODUCTION

Almost all societies around the world face the challenges of aging population with increasing numbers of older adults. An important cognitive ability for successful aging is Prospective memory (PM). PM refers to our ability to remember and execute tasks at a specified time or in relation to another event in the future [1]. For older adults, intact PM becomes essential to retain functional independence and maintain overall well-being.

However, despite its importance, current clinical tools for PM assessment lack ecological validity. A widely used paradigm for laboratory assessment of PM is Virtual Week [2]. Virtual Week is a digital board game that simulates an everyday-like week, with one board round representing one day. Participants roll a die to go around, draw event cards (e.g., having breakfast, doing laundry, etc.), and must remember to perform everyday intentions (e.g., taking medication, returning a book at the library) that are either linked to the events or certain moments in time.

While Virtual Week is widely adopted in clinical and non-clinical settings, it is still a board game that lacks an everyday-like experience. Alternatively, Virtual Reality (VR) offers a 3D environment with realistic scenes and life-size objects, making the user feel present and immersed during the interaction. Such immersive environments can elicit psychological and physical responses akin to those experienced in real-life situations making them appropriate for assessing goal-directed behavior and PM [9].

Building on this premise, we introduce “Virtual Day” as a new VR game for the assessment of PM based on the well-established “Virtual Week” paradigm. “Virtual Day” seeks to create a reliable and standardized VR environment as an ecologically valid platform for not only assessing but also potentially training PM in older adults. This paper presents the development and validation of the first viable prototype of Virtual Day.

2 METHODOLOGY

2.1 Participants

A total of 24 participants (18 females, 6 males, $M_{age} = 24.08$, $SD_{age} = 4.55$) who had limited or no prior experience with VR were recruited. The study was approved by the Ethics Review Board of Tilburg School of Social and Behavioral Sciences.

2.2 The VR game

The development of the VR game “Virtual Day” was inspired by the tasks and elements already implemented in the Virtual Week paradigm [1, 2]. Unlike Virtual Week, which is a digital board game, in Virtual Day, the tasks require navigating through 3D spaces in the virtual town, for instance, taking medication from the medicine cabinet in the bathroom, watering plants in the living room, dropping a parcel at a post box, etc.). The game allows players to grab objects in the environment using VR controllers and move around using teleport rays. Every time the player completes one task, the time in the game advances quickly to the next task of the day. Similar to Virtual Week, the tasks are either time-based (e.g., taking medication at a specific time) or event-based (e.g., buying items when they appear). PM performance is measured by computing the percentage of completed tasks overall, as well as by the type of the task (time vs. event-based).

2.3 Procedure

Upon providing consent, participants completed a demographics and the Prospective-Retrospective Memory Questionnaire (PRMQ) [3]. The PRMQ is an effective instrument for measuring the self-perceived frequency of both prospective and retrospective memory failures. Next, participants completed the two experimental conditions (Virtual Week, hereafter referred to as the digital board game, and Virtual Day, hereafter referred to as the VR game) in a randomized order. During the digital board game, participants were seated behind a computer, whereas for the VR game, participants wore the Oculus Rift S headset. Both conditions entailed a familiarization phase at the beginning during which participants practiced the game. Next participants entered the evaluation phase in which they conducted the planned activities for PM evaluation.



Figure 1: Scenes from different spaces and objects in the Virtual Day game, including A) the house in which the player lived B) the supermarket for grocery shopping, C) plants for the watering task, D) an inhaler used as medication, and E) a postbox outside of the house for delivery of parcels.

During the evaluation phase, in addition to the planned task, players also received texts with new unplanned tasks as event-based tasks (e.g., dropping off a parcel). In the VR game, these unplanned tasks showed up as tasks associated with existing objects in the environment. In the digital board game, participants received a message to draw a card for an additional unplanned task. Each condition took approximately 20 minutes, following which participants filled out the UXIVE questionnaire (see 2.4). Finally, participants were debriefed and the experiment ended.

2.4 Evaluation

We utilized the UXIVE questionnaire [4] to capture the multi-faceted user experience within virtual environments using nine constructs; Presence, Engagement, Immersion, Flow, Skill, Emotion, Technology Adoption, Judgment, and Experience Consequence (see Supplementary Materials for more details). Participants' responses were captured using a 5-point Likert scale, except for Judgment, which utilizes binary answers. To evaluate user experience, the mean scores from the VR and the board game were compared across all UXIVE constructs.

To evaluate PM performance, this study aligned the PRMQ results with the scores obtained from both VR and digital board games. Specifically, we examined the correlation between the overall performance, represented by the percentage of completed tasks, as well as the event and time-based task scores with the outcomes of the PRMQ.

3 RESULTS

3.1 User Experience

Figure 2 demonstrates the outcome of the UXIVE questionnaire for both games per each construct. A significant difference was found for Engagement $t(23) = 2.36, p = .027$, Immersion $t(23) = 4.17, p < .001$, Flow $t(23) = 2.21, p = .038$ and Emotion $t(23) = 2.80, p = .010$, indicating a higher score for the VR game compared to the digital board game. On the other hand, metrics related to ease of use, such as Skill and Experience Consequence showed an opposite pattern; the digital board game required significantly less Skill $t(23) = -2.36, p = .027$ and had lower Experience consequences $t(23) = -2.92, p = .008$ compared to the VR game. No significant differences were found for Technology Adoption and Judgment between the VR and board games.

3.2 Prospective Memory Performance

To understand the relationships between different task performances and PRMQ results, we employed Spearman's rank correlation analysis. Analyzing the total PM scores for all tasks, we observed a small positive association between the two games. However, this correlation was not statistically significant, with $\rho(24) = 0.35, p = 0.153$. Importantly, the all-tasks scores from both Virtual Day and Virtual Week revealed weak correlations with

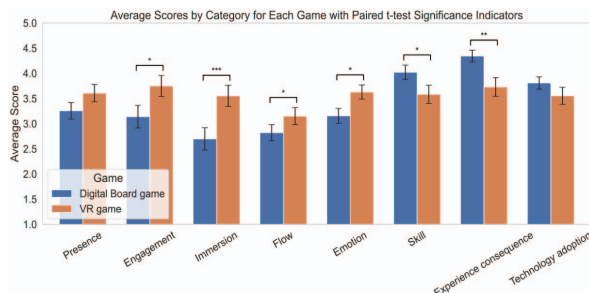


Figure 2: Comparison of UXIVE constructs between the two games (* $p < .05$, ** $p < .01$, *** $p < .001$).

PRMQ results. This suggests a marked discrepancy between participants' self-assessed PM and their actual performance in virtual environments.

Focusing on the two types of tasks within the games, we found that the correlation between event-based tasks of Virtual Day and Virtual Week was moderate and statistically significant, with $\rho(24) = 0.51, p = 0.030$, showing a certain degree of coherence in event-based task performance across the two platforms. However, the relationship between time-based tasks, while being positive, was non-significant ($p = 0.322$), indicating inconsistent performance in time-based tasks between the Virtual Day and Virtual Week.

4 DISCUSSION

The present study introduced a new VR game, Virtual Day, as a more accurate and ecologically valid tool for assessing prospective memory in older adults. In a pilot study with a group of young adults, we compared user experience with Virtual Day as opposed to the state-of-the-art tool for PM assessment which is a digital board game. Participants reported their experience with Virtual Day as significantly more immersive and engaging, feeling a heightened state of flow (being "in the zone") and a richer emotional depth compared to the traditional 2D computerized board game. This underscores the potential of VR environments to offer a more captivating and enjoyable experience to users.

Conversely, the board game received higher scores for the Skill and Experience Consequence dimensions which represent users' confidence and potential negative symptoms, like cybersickness, that users might encounter in a virtual environment. This trend was expected due to the novelty of VR for most participants [13]. Future studies could benefit from a longer familiarization phase to help users become more comfortable navigating with VR controllers.

When examining the PM task performance in Virtual Day and Virtual Week games, specific correlation patterns were observed across different task types. While no significant correlations were found between all-task scores or time-based task scores of the two games, a moderate and significant correlation was noted for event-based tasks. Virtual Day's all-tasks scores showed a higher association with PRMQ scores compared to Virtual Week, suggesting a potential for Virtual Day to bridge the gap between lab assessments and real-life PM performance due to its higher ecological validity.

Building upon the results of the current feasibility study, we plan to expand the VR game by increasing the number of tasks, allowing for multiple scenarios and variations, similar to Virtual Week. Additionally, we plan to conduct a pilot study with older adults to further examine the feasibility of the new VR tool for the target end users. Feedback from older participants will be instrumental in refining the PM tasks, ensuring they remain realistic and user-friendly. Finally, beyond serving as an assessment tool for PM performance, we aim to investigate the possibilities of Virtual Day as a medium to enhance cognitive functions in older adults.

REFERENCES

- [1] Hering, A., Kliegel, M., Rendell, P. G., Craik, F. I., & Rose, N. S. (2018). Prospective memory is a key predictor of functional independence in older adults. *Journal of the International Neuropsychological Society*, 24(6), 640-645.
- [2] Rendell, P. G., & Henry, J. D. (2009). A review of Virtual Week for prospective memory assessment: Clinical implications. *Brain Impairment*, 10(1), 14-22.
- [3] Smith, G., Del Sala, S., Logie, R. H., & Maylor, E. A. (2000). Prospective and retrospective memory in normal ageing and dementia: A questionnaire study. *Memory*, 8(5), 311-321.
- [4] Tcha-Tokey, K., Christmann, O., Loup-Escande, E., & Richir, S. (2016). Proposition and validation of a questionnaire to measure the user experience in immersive virtual environments. *International Journal of Virtual Reality*, 16(1), 33-48.